LAPORAN PRAKTIKUM

POSTTEST 7

JARINGAN KOMPUTER



Disusun oleh:

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Kelas (B ‘25)

PROGRAM STUDI INFORMATIKA

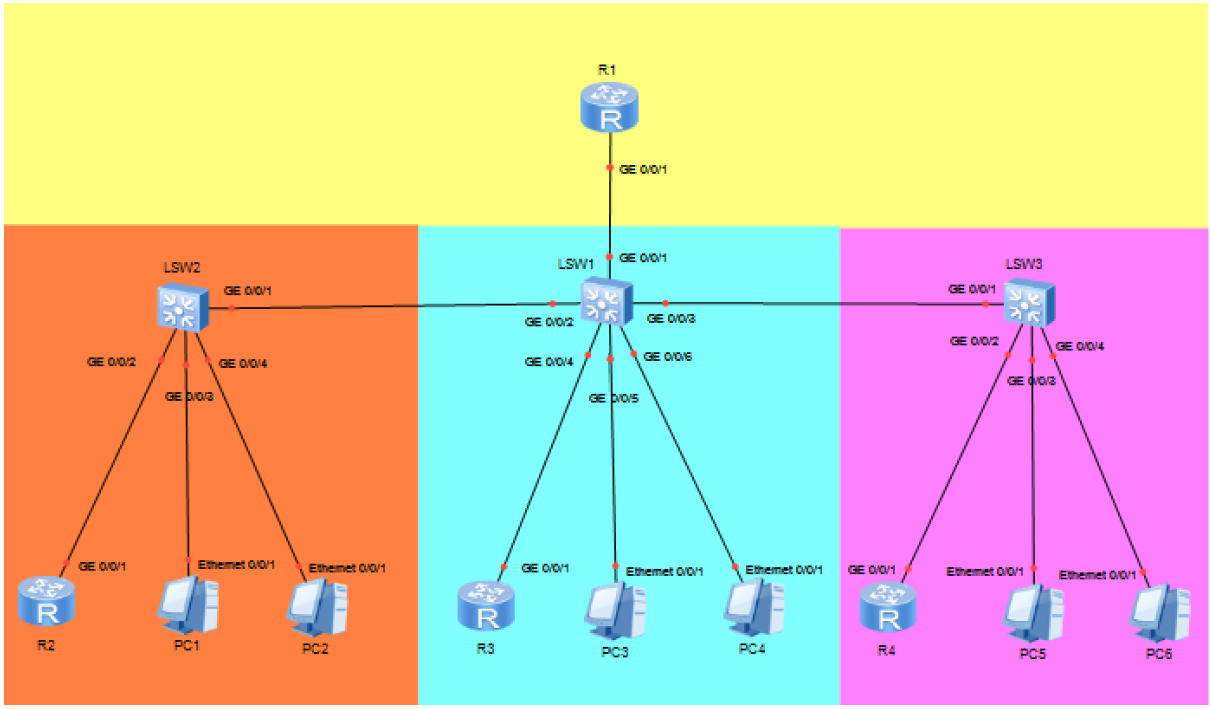
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SAMARINDA

2026

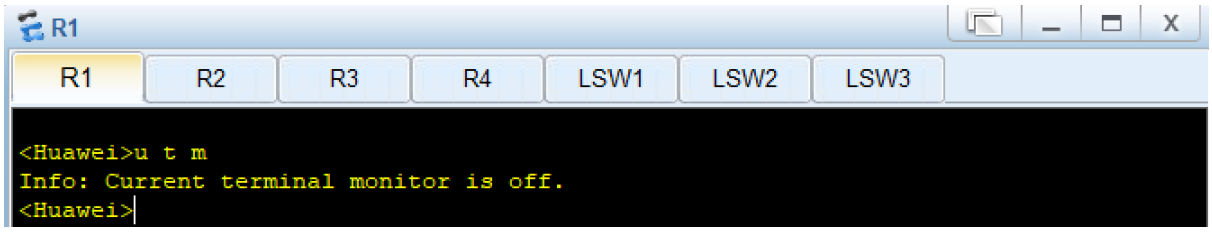
1. Soal 1

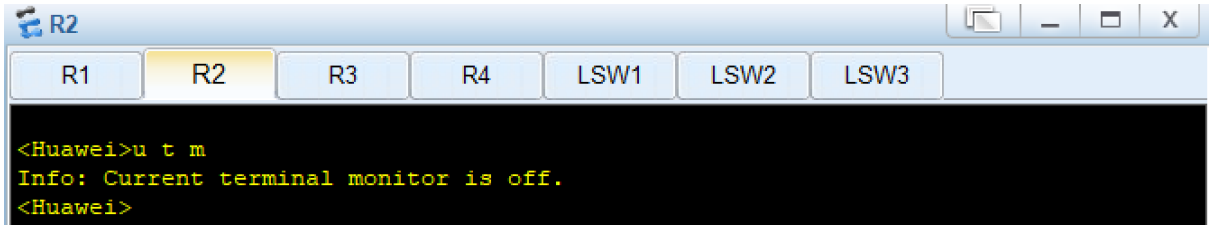
Langkah pertama adalah membangun topologi fisik yang terdiri dari 4 buah router, 3 buah switch, dan 6 buah PC. Pasang kabel copper dari masing-masing perangkat ke perangkat lain dengan interface yang sesuai topologi pada posttest. Mulai topologi jaringan dengan menekan tombol start di bagian atas fitur pada ensp.

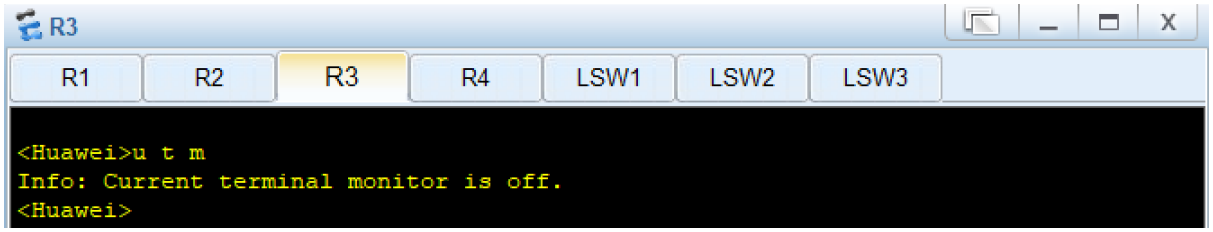


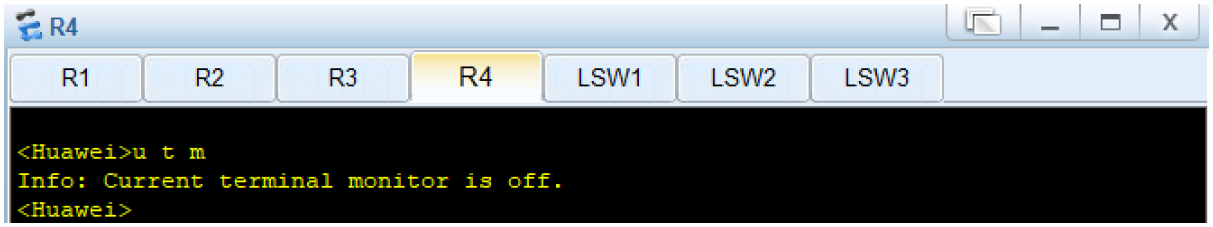
2. Soal 2

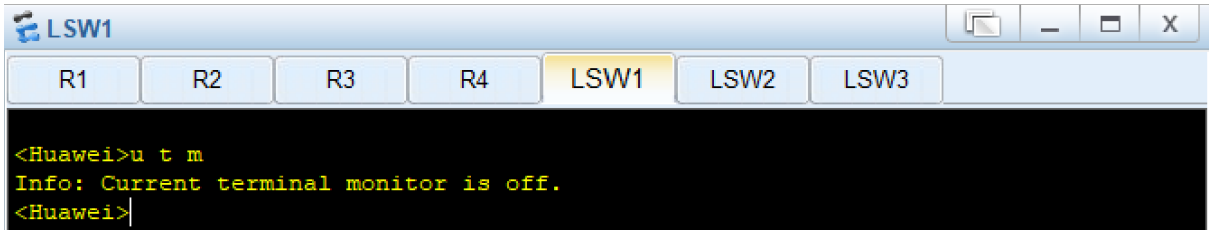
Ketik u t m untuk menghilangkan log panjang yang akan muncul seiring melakukan konfigurasi. Lakukan hal ini untuk semua perangkat router dan switch (R1, R2, R3, R4, LSW1, LSW2, dan LSW3).

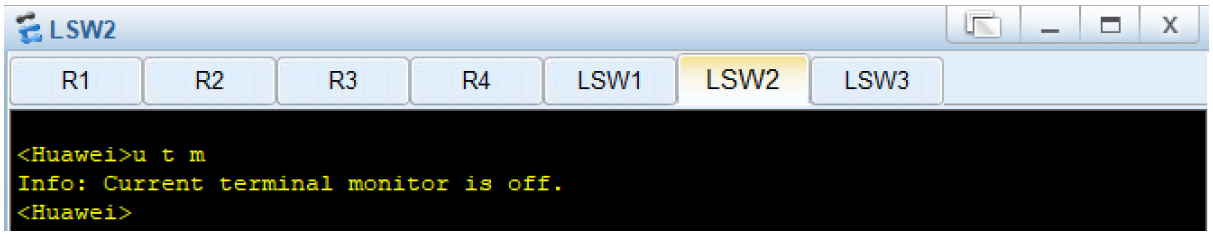


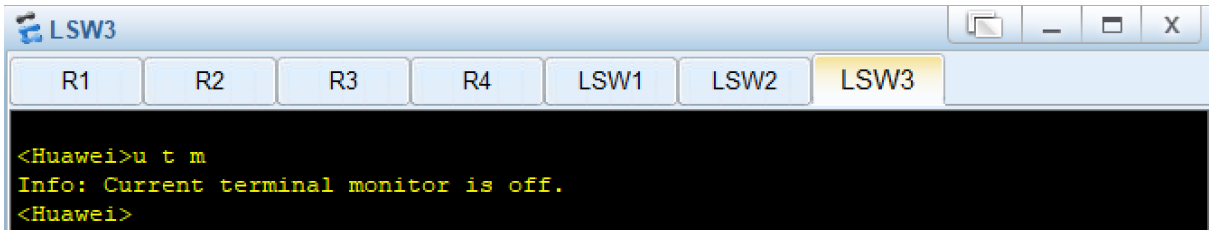












3. Soal 3

Untuk mengganti nama router dan switch sesuai ketentuan pada soal di classroom (R*_SERVER_067, S*_067, R*_CLIENT_067), konfigurasi dilakukan melalui CLI pada masing-masing router dan switch. Masuk ke system-view, lalu sysname R1_SERVER_067. Lakukan hal yang sama kepada setiap router dan switch.

R1

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1_SERVER_067
[R1_SERVER_067]
```

R2

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2_CLIENT_067
[R2_CLIENT_067]
```

R3

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3_CLIENT_067
[R3_CLIENT_067]
```

R4

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4_CLIENT_067
[R4_CLIENT_067]
```

LSW1

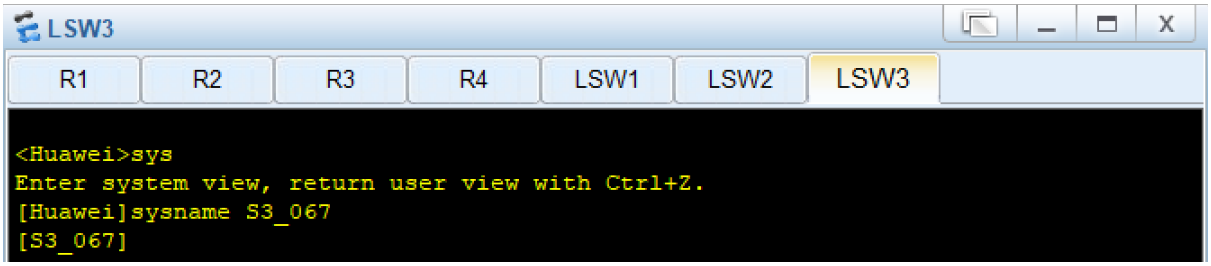
R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1_067
[S1_067]
```

LSW2

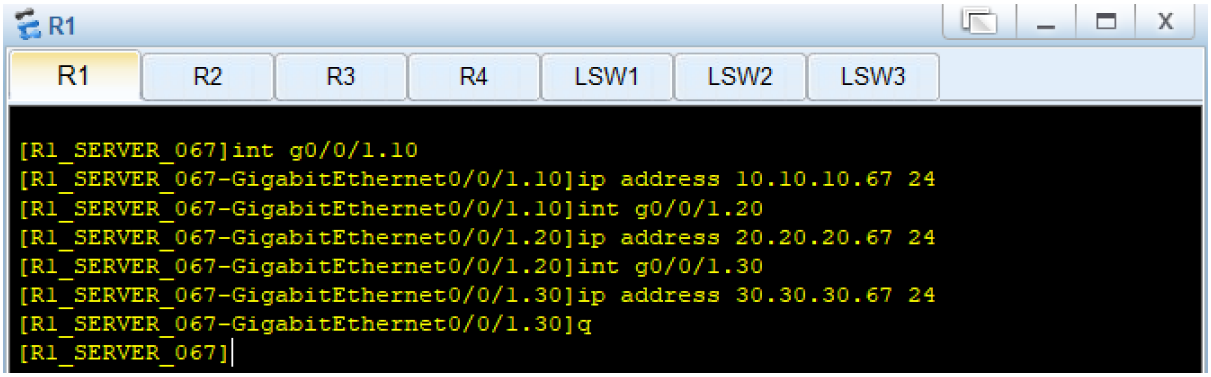
R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
<Huawei>sys
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2_067
[S2_067]
```



4. Soal 4

Konfigurasikan interface dasar, meliputi pembuatan sub-interface dan ip ip statis pada R1, penetapan aturan trunk/access VLAN di setiap port switch, serta mengaktifkan mode dhcp, pendefinisian mode ip dinamis (ip address dhcp-alloc) pada interface router client maupun PC.



LSW1

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
[S1_067]vlan batch 10 20 30
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1_067]int g0/0/1
[S1_067-GigabitEthernet0/0/1]port link-type trunk
[S1_067-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20 30
[S1_067-GigabitEthernet0/0/1]int g0/0/2
[S1_067-GigabitEthernet0/0/2]port link-type trunk
[S1_067-GigabitEthernet0/0/2]port trunk allow-pass vlan 10 20 30
[S1_067-GigabitEthernet0/0/2]int g0/03
                             ^
Error: Unrecognized command found at '^' position.
[S1_067-GigabitEthernet0/0/2]int g0/0/3
[S1_067-GigabitEthernet0/0/3]port link-type trunk
[S1_067-GigabitEthernet0/0/3]port trunk allow-pass vlan 10 20 30
[S1_067-GigabitEthernet0/0/3]int g0/0/4
[S1_067-GigabitEthernet0/0/4]port link-type access
[S1_067-GigabitEthernet0/0/4]port default vlan 10
[S1_067-GigabitEthernet0/0/4]int g0/0/5
[S1_067-GigabitEthernet0/0/5]port link-type access
[S1_067-GigabitEthernet0/0/5]port default vlan 20
[S1_067-GigabitEthernet0/0/5]int g0/0/6
[S1_067-GigabitEthernet0/0/6]port link-type access
[S1_067-GigabitEthernet0/0/6]port default vlan 30
[S1_067-GigabitEthernet0/0/6]q
[S1_067]
```

LSW2

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
[S2_067]vlan batch 10 20 30
Info: This operation may take a few seconds. Please wait for a moment...done.
[S2_067]int g0/0/1
[S2_067-GigabitEthernet0/0/1]port link-type trunk
[S2_067-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20 30
[S2_067-GigabitEthernet0/0/1]int g0/0/2
[S2_067-GigabitEthernet0/0/2]port link-type access
[S2_067-GigabitEthernet0/0/2]port default vlan 10
[S2_067-GigabitEthernet0/0/2]int g0/0/3
[S2_067-GigabitEthernet0/0/3]port link-type access
[S2_067-GigabitEthernet0/0/3]port default vlan 20
[S2_067-GigabitEthernet0/0/3]int g0/0/4
[S2_067-GigabitEthernet0/0/4]port link-type access
[S2_067-GigabitEthernet0/0/4]port default vlan 30
[S2_067-GigabitEthernet0/0/4]q
[S2_067]
```

LSW3

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
[S3_067]vlan batch 10 20 30
Info: This operation may take a few seconds. Please wait for a moment...done.
[S3_067]int g0/0/1
[S3_067-GigabitEthernet0/0/1]port link-type trunk
[S3_067-GigabitEthernet0/0/1]port trunk allow-pass vlan 10 20 30
[S3_067-GigabitEthernet0/0/1]int g0/0/2
[S3_067-GigabitEthernet0/0/2]port link-type access
[S3_067-GigabitEthernet0/0/2]port default vlan 10
[S3_067-GigabitEthernet0/0/2]int g0/0/3
[S3_067-GigabitEthernet0/0/3]port link-type access
[S3_067-GigabitEthernet0/0/3]port default vlan 20
[S3_067-GigabitEthernet0/0/3]int g0/0/4
[S3_067-GigabitEthernet0/0/4]port link-type access
[S3_067-GigabitEthernet0/0/4]port default vlan 30
[S3_067-GigabitEthernet0/0/4]q
[S3_067]
```

R2

R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

```
[R2_CLIENT_067]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R2_CLIENT_067]int g0/0/1
[R2_CLIENT_067-GigabitEthernet0/0/1]ip address dhcp-alloc
[R2_CLIENT_067-GigabitEthernet0/0/1]q
[R2_CLIENT_067]
```

R3

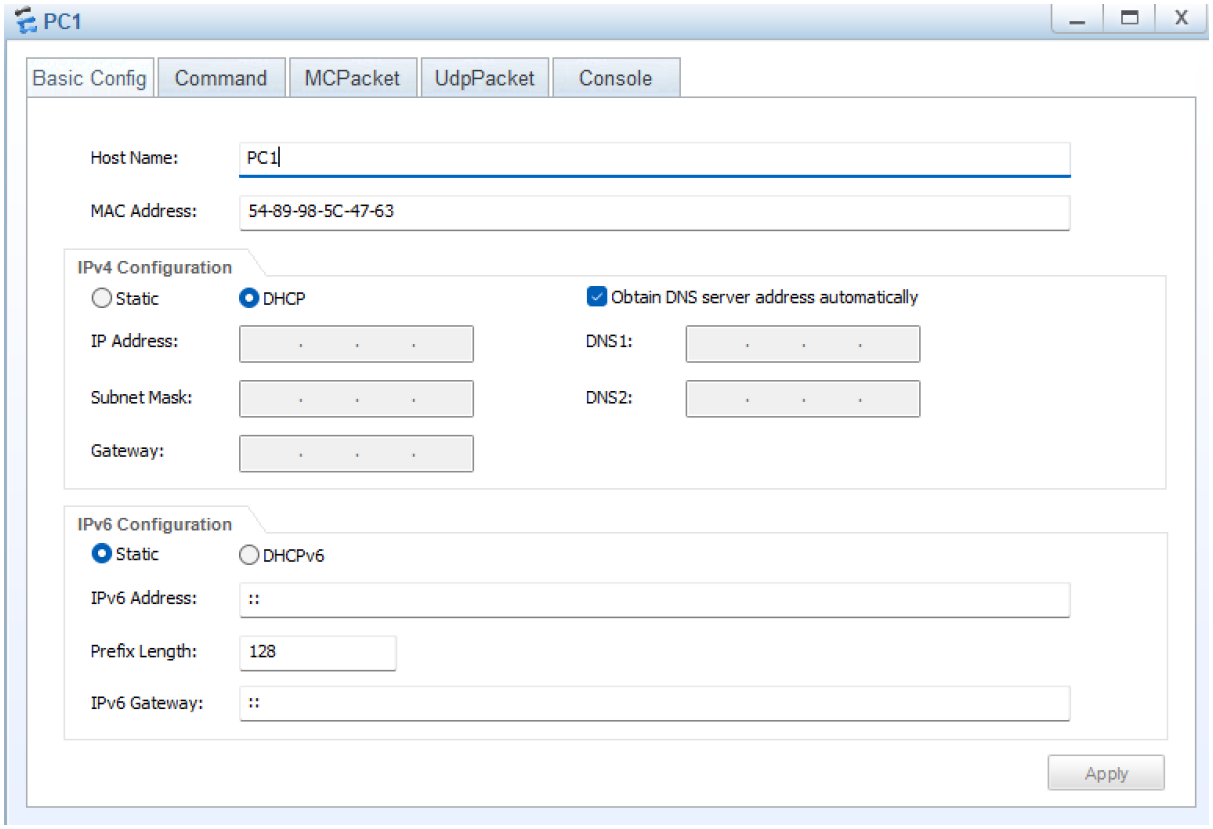
R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

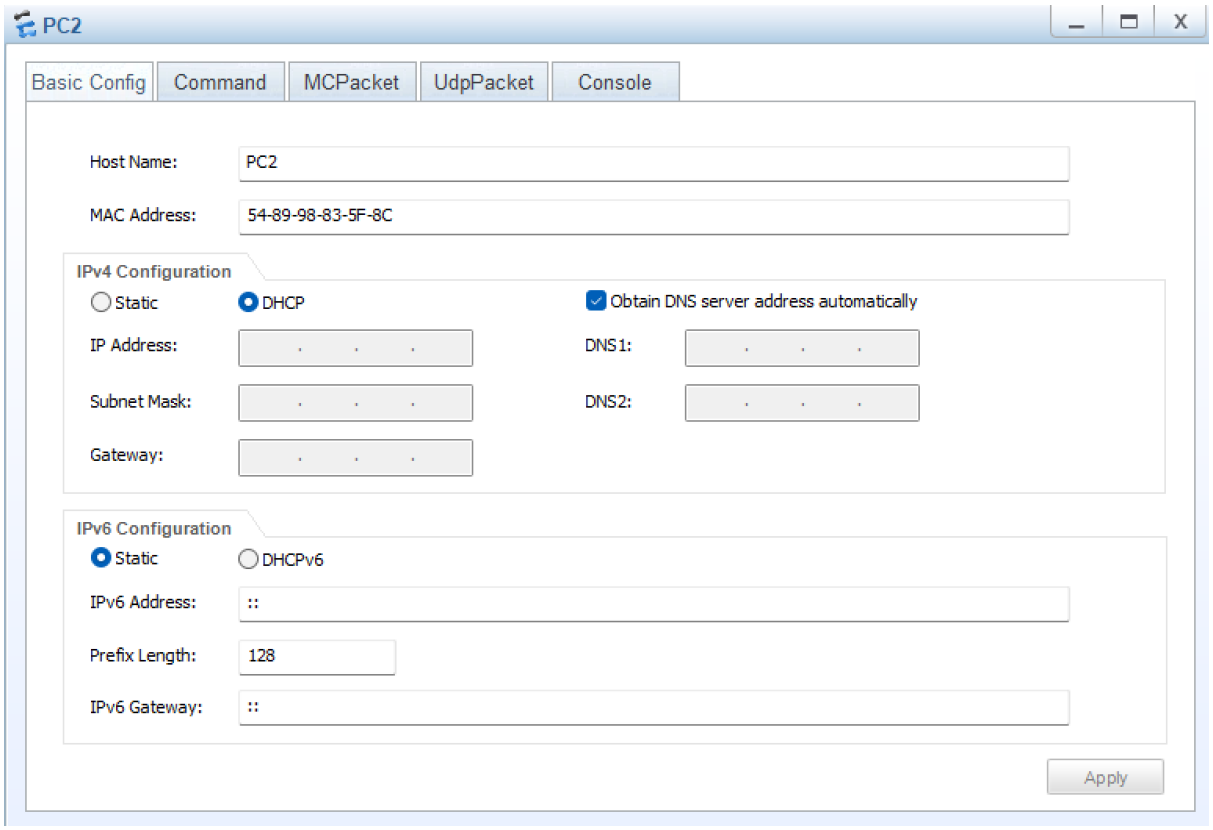
```
[R3_CLIENT_067]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R3_CLIENT_067]int g0/0/1
[R3_CLIENT_067-GigabitEthernet0/0/1]ip address dhcp-alloc
[R3_CLIENT_067-GigabitEthernet0/0/1]q
[R3_CLIENT_067]
```

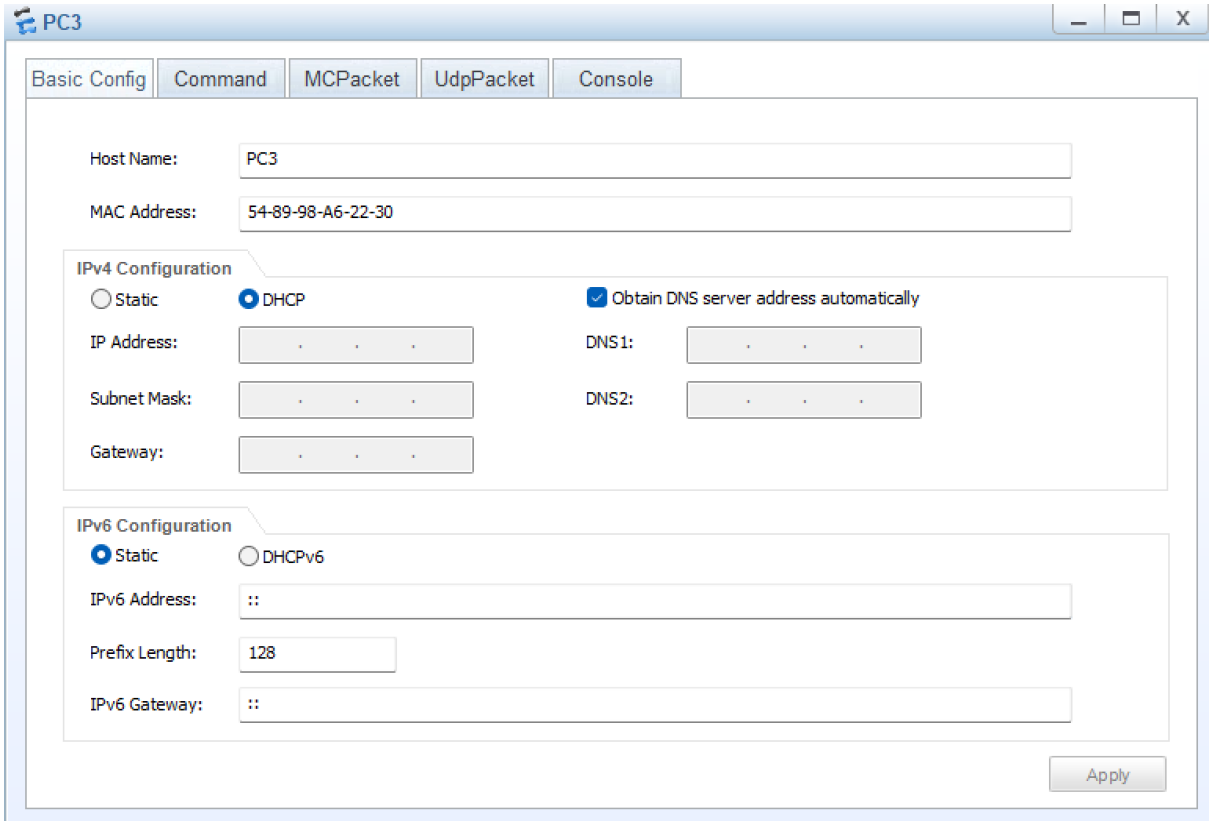
R4

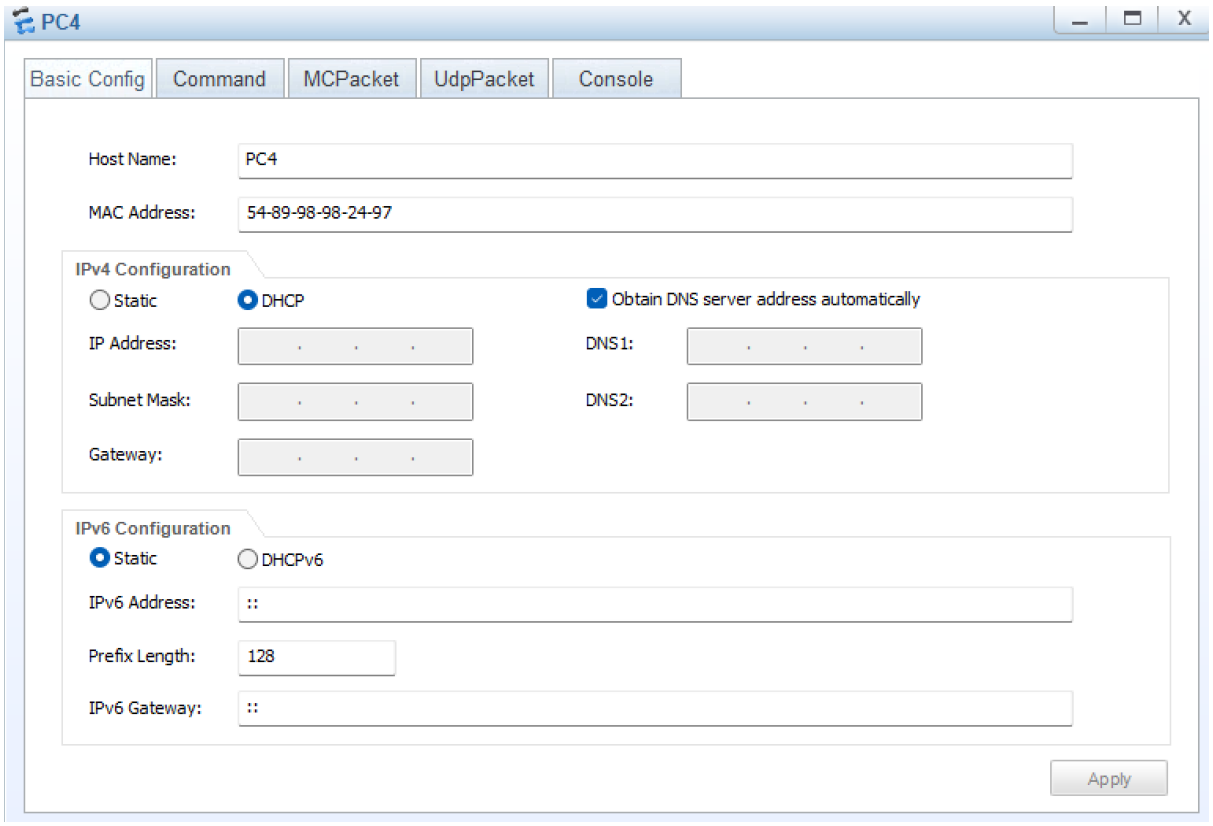
R1 | R2 | R3 | R4 | LSW1 | LSW2 | LSW3

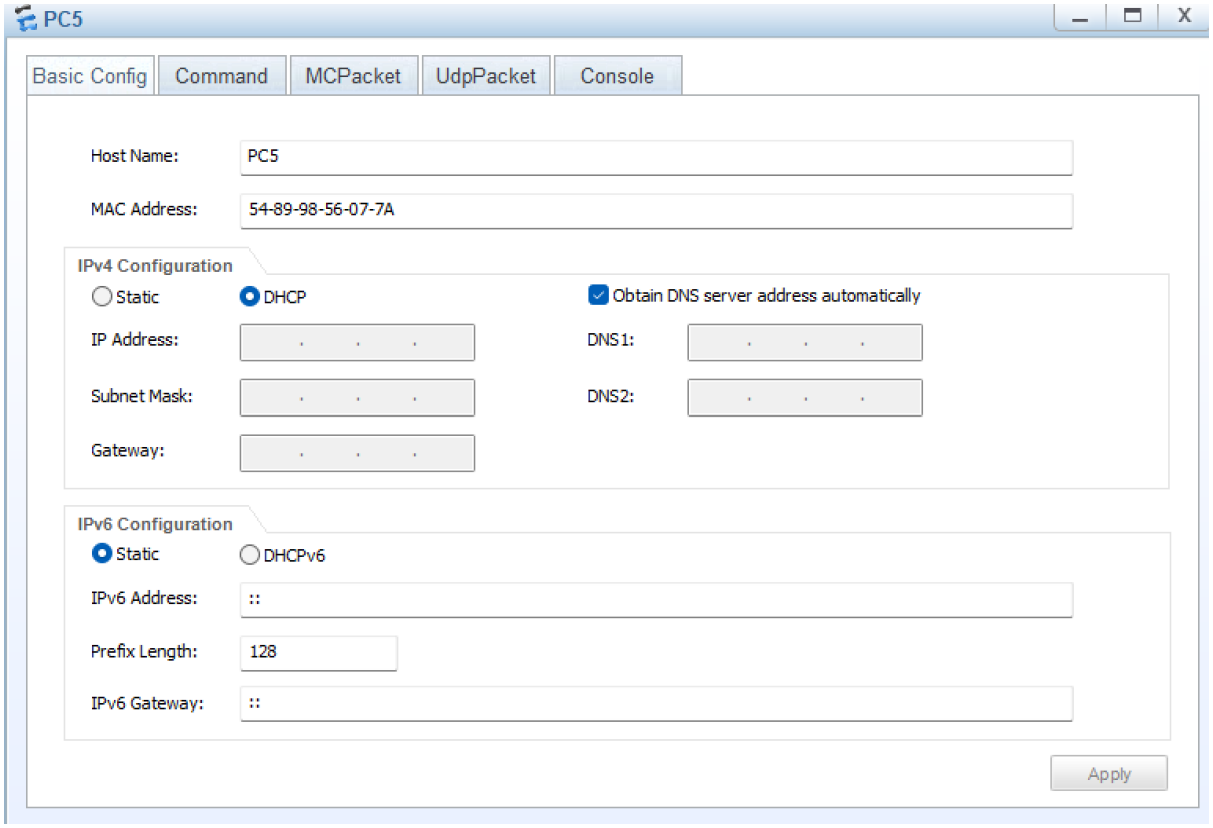
```
[R4_CLIENT_067]dhcp enable
Info: The operation may take a few seconds. Please wait for a moment.done.
[R4_CLIENT_067]int g0/0/1
[R4_CLIENT_067-GigabitEthernet0/0/1]ip address dhcp-alloc
[R4_CLIENT_067-GigabitEthernet0/0/1]q
[R4_CLIENT_067]
```

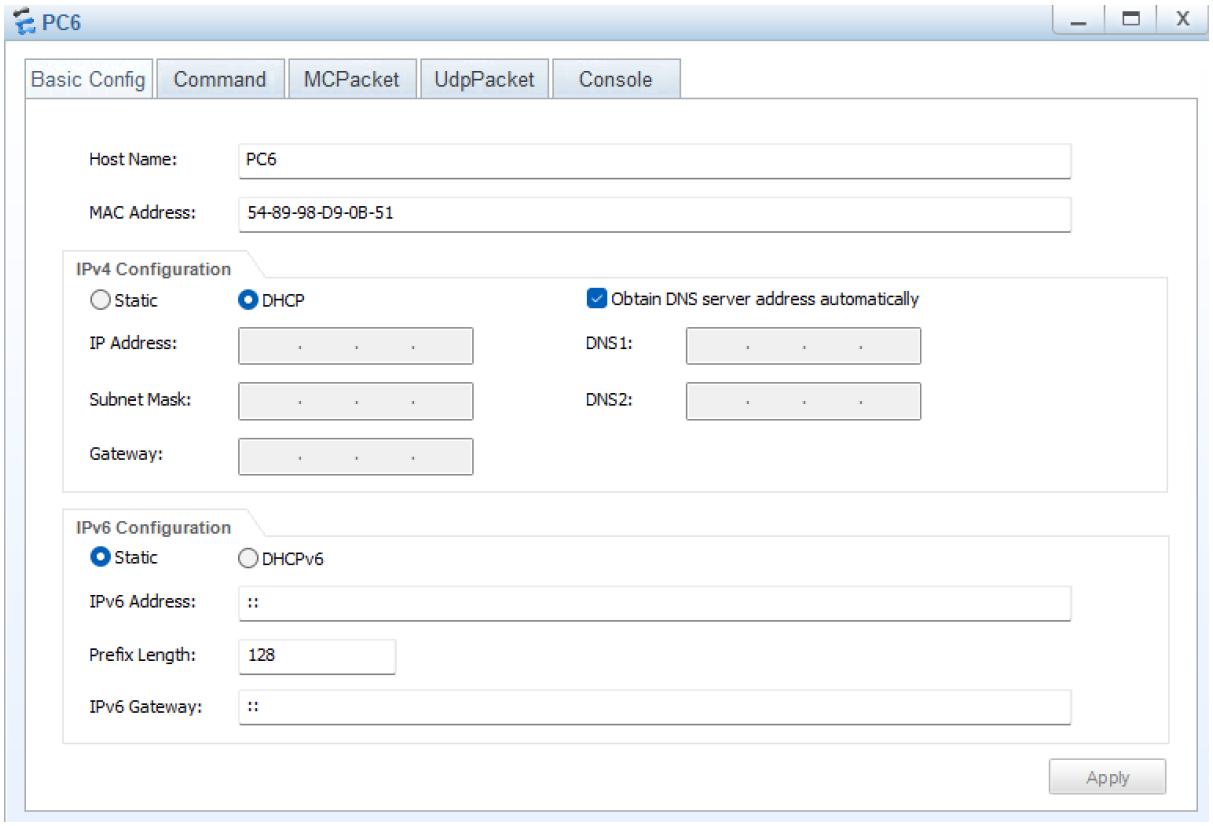

PC1
Basic Config
Command
MCPacket
UdpPacket
Console
Host Name:
PC1
MAC Address:
54-89-98-5C-47-63
IPv4 Configuration
Static
DHCP
Obtain DNS server address automatically
IP Address:
DNS1:
Subnet Mask:
DNS2:
Gateway:
IPv6 Configuration
Static
DHCPv6
IPv6 Address:
::
Prefix Length:
128
IPv6 Gateway:
::
Apply


PC2
Basic Config
Command
MCPacket
UdpPacket
Console
Host Name:
PC2
MAC Address:
54-89-98-83-5F-8C
IPv4 Configuration
Static
DHCP
Obtain DNS server address automatically
IP Address:
DNS1:
Subnet Mask:
DNS2:
Gateway:
IPv6 Configuration
Static
DHCPv6
IPv6 Address:
::
Prefix Length:
128
IPv6 Gateway:
::
Apply


PC3
Basic Config
Command
MCPacket
UdpPacket
Console
Host Name:
PC3
MAC Address:
54-89-98-A6-22-30
IPv4 Configuration
Static
DHCP
Obtain DNS server address automatically
IP Address:
DNS1:
Subnet Mask:
DNS2:
Gateway:
IPv6 Configuration
Static
DHCPv6
IPv6 Address:
::
Prefix Length:
128
IPv6 Gateway:
::
Apply

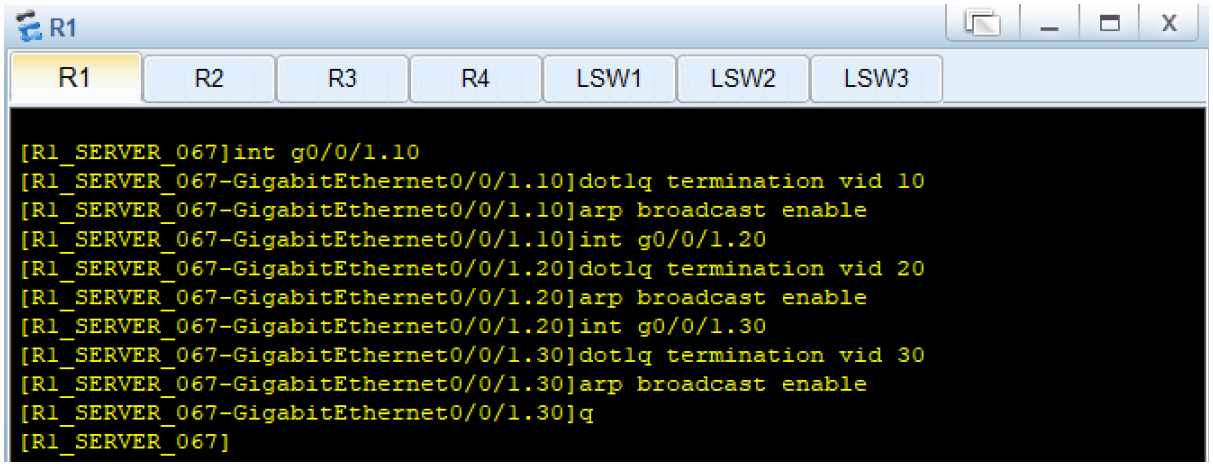

PC4
Basic Config
Command
MCPacket
UdpPacket
Console
Host Name:
PC4
MAC Address:
54-89-98-98-24-97
IPv4 Configuration
Static
DHCP
Obtain DNS server address automatically
IP Address:
DNS1:
Subnet Mask:
DNS2:
Gateway:
IPv6 Configuration
Static
DHCPv6
IPv6 Address:
::
Prefix Length:
128
IPv6 Gateway:
::
Apply


PC5
Basic Config
Command
MCPacket
UdpPacket
Console
Host Name:
PC5
MAC Address:
54-89-98-56-07-7A
IPv4 Configuration
Static
DHCP
Obtain DNS server address automatically
IP Address:
DNS1:
Subnet Mask:
DNS2:
Gateway:
IPv6 Configuration
Static
DHCPv6
IPv6 Address:
::
Prefix Length:
128
IPv6 Gateway:
::
Apply


PC6
Basic Config
Command
MCPacket
UdpPacket
Console
Host Name:
PC6
MAC Address:
54-89-98-D9-0B-51
IPv4 Configuration
Static
DHCP
Obtain DNS server address automatically
IP Address:
DNS1:
Subnet Mask:
DNS2:
Gateway:
IPv6 Configuration
Static
DHCPv6
IPv6 Address:
::
Prefix Length:
128
IPv6 Gateway:
::
Apply

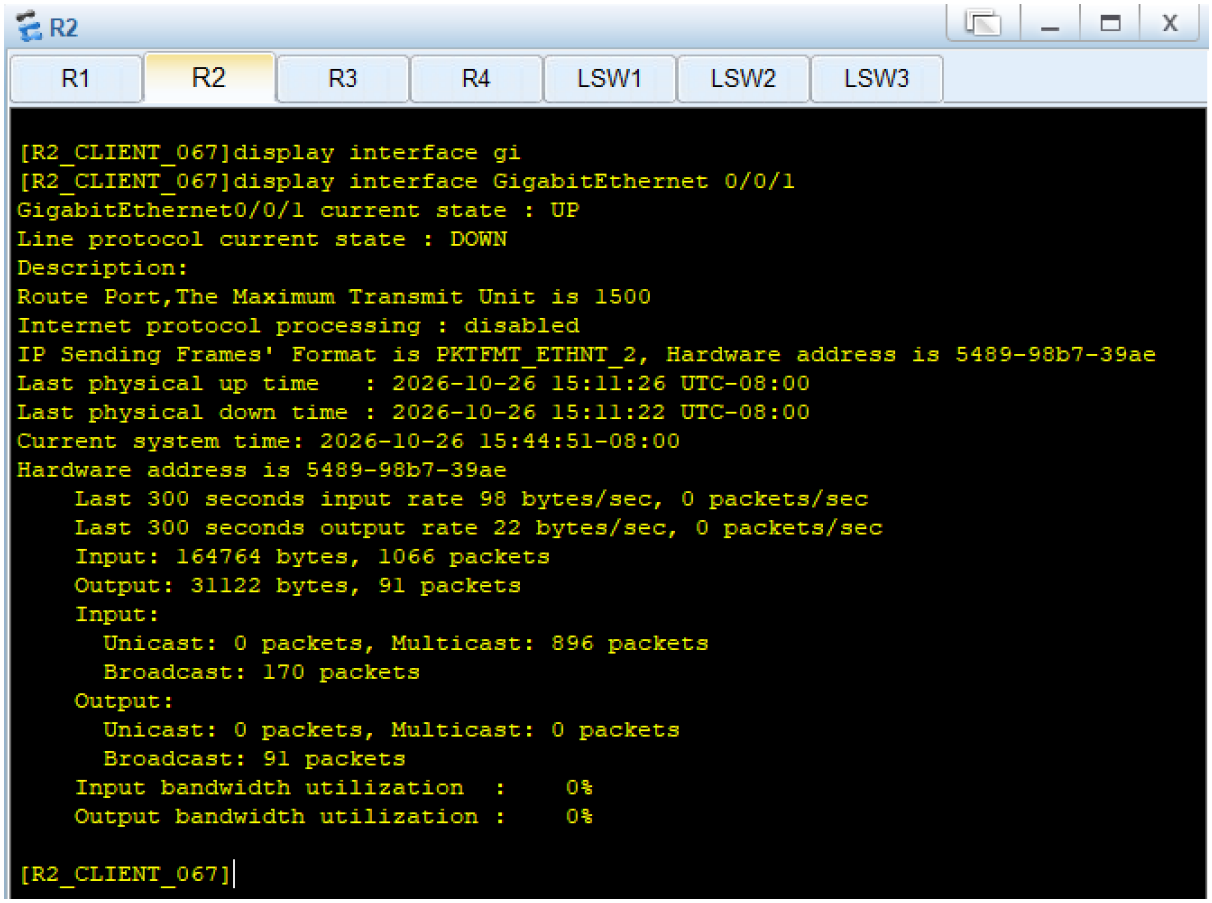
5. Soal 5

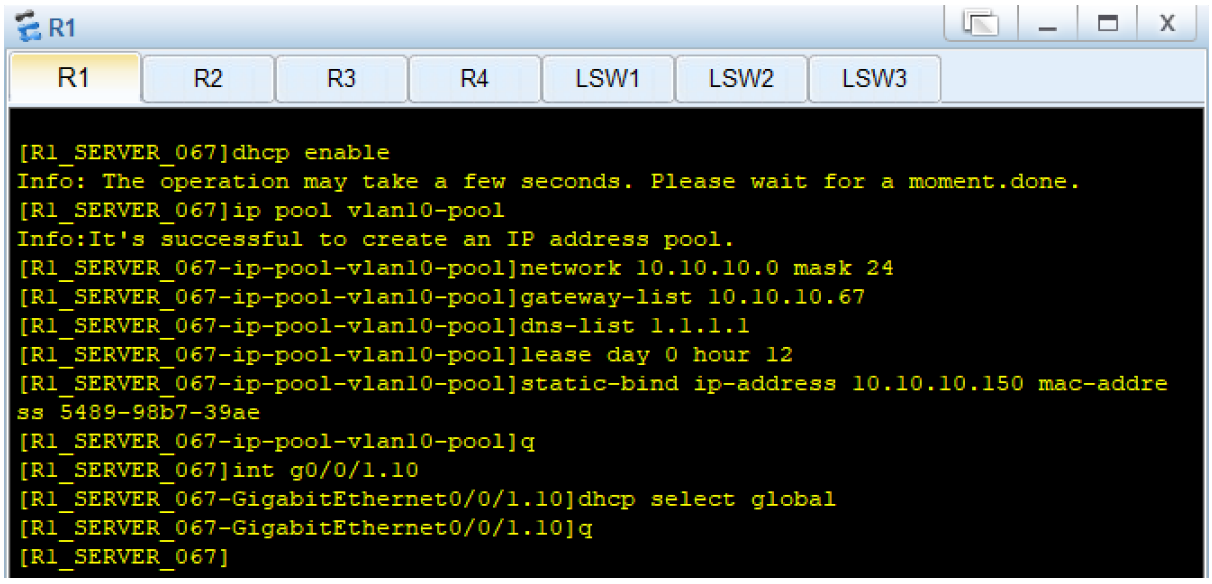
Gunakan inter VLAN untuk menghubungkan setiap VLAN dengan command dot1q termination pada sub-interface R1 agar router sanggup memproses tagging dari VLAN ID 10, 20, dan 30. Lalu, eksekusi arp broadcast enable agar gateway dapat merespons req dari ARP dari perangkat end-device.



6. Soal 6

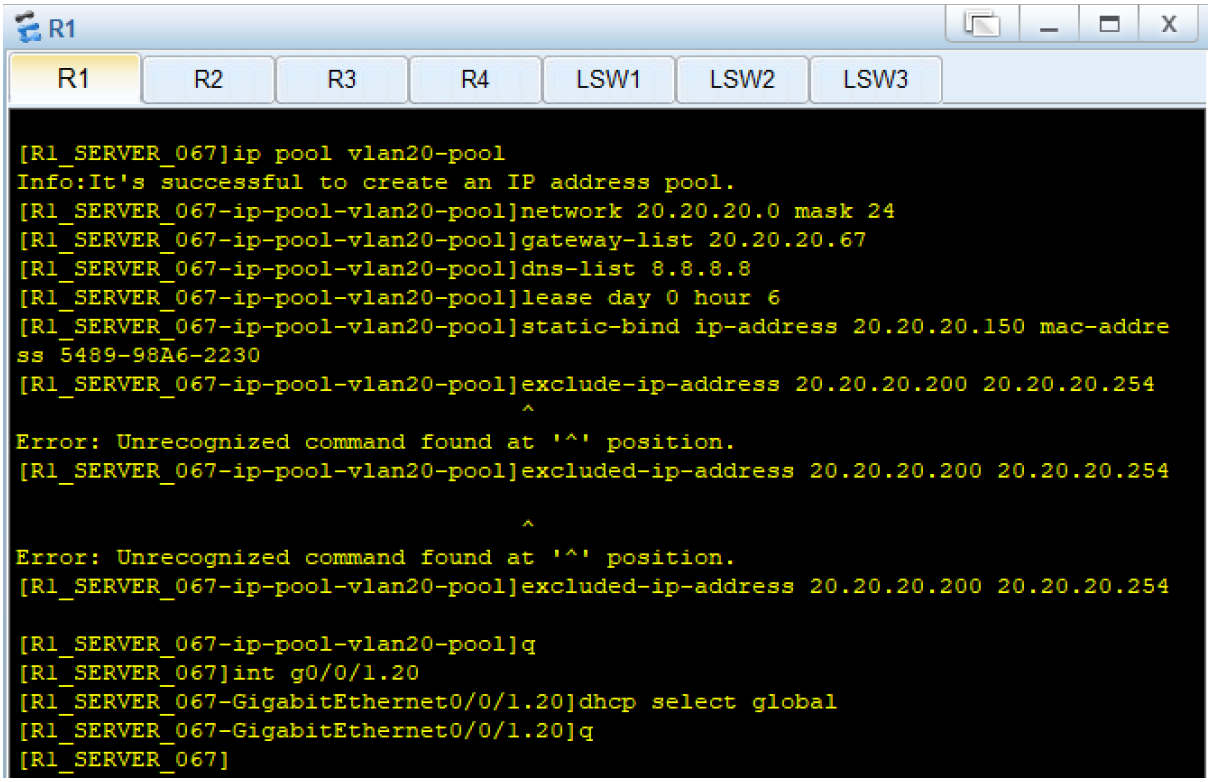
Mengecek MAC Address R2 untuk interface ge0/0/1 dari perintah display interface GigabitEthernet 0/0/1. Dari display, kita tahu jika MAC Address R2 adalah 5489-98b7-39ae. Konfigurasikan DHCP global pool vlan10-pool di R1 untuk mengelola req jaringan 10.10.10.0 /24. Ketentuan konfigurasi tambahan, yaitu DNS 1.1.1.1, masa sewa 12 jam, dan bind static ip untuk perangkat dengan MAC Address seperti di gambar (R2). Lalu, terapkan konfigurasi ip pool tadi ke dalam sub-interface ge0/0/1.10 dengan dhcp select global.





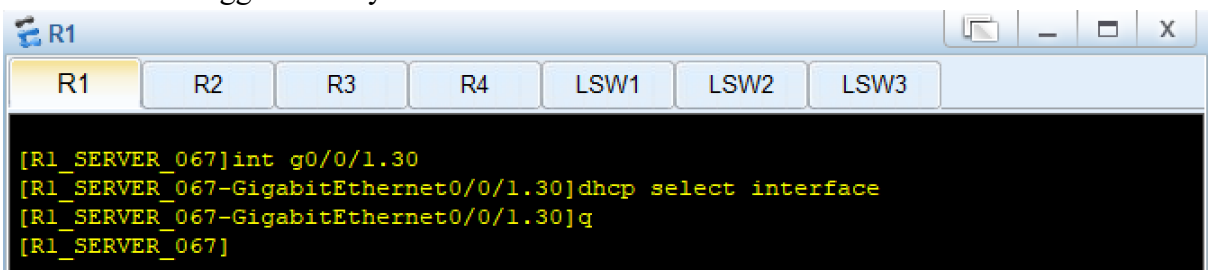
7. Soal 7

Konfigurasikan DHCP global pool vlan20-pool di R1 untuk mengelola req ip jaringan 20.20.20.0 /24. Ketentuan konfigurasi tambahan, yaitu DNS 8.8.8.8, masa sewa 6 jam, bind ip static untuk perangkat dengan MAC Address seperti di gambar (PC 3), serta pengecualian ip yang tidak boleh diberikan dari rentang ip 20.20.20.200 hingga 20.20.20.254. Lalu, terapkan konfigurasi ip pool tadi ke dalam sub-interface ge0/0/1.20 dengan dhcp select global.



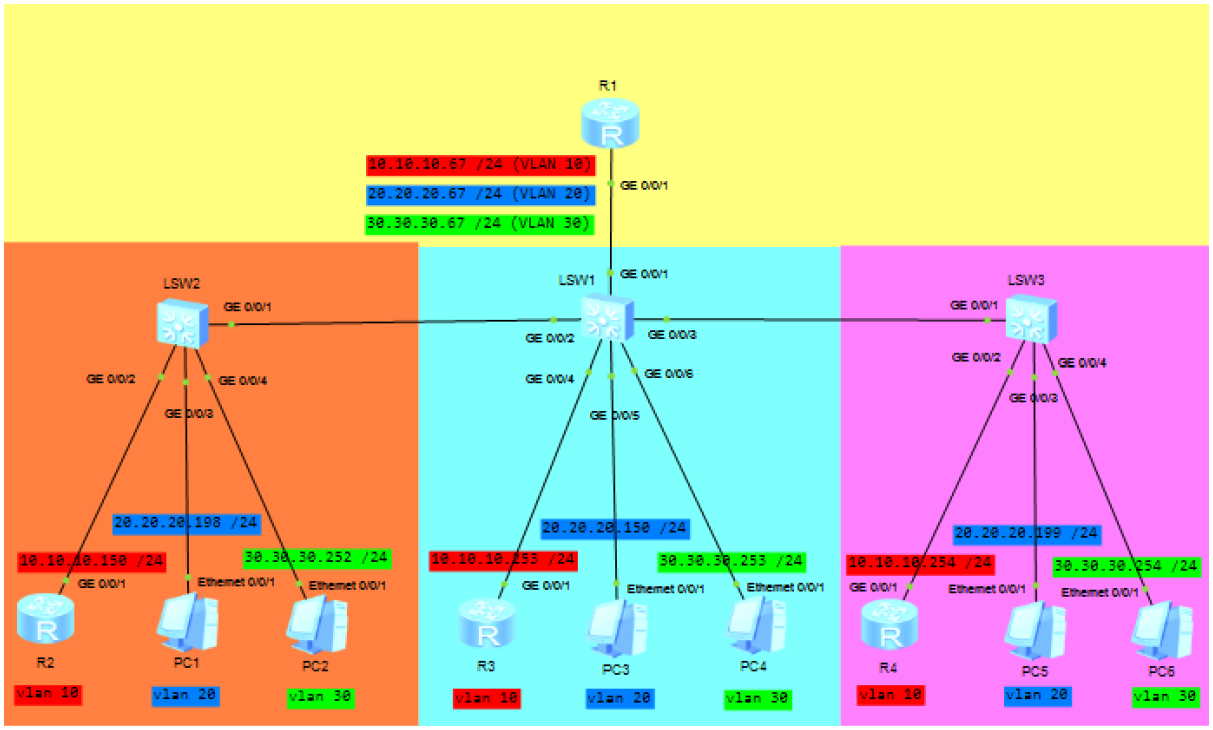
8. Soal 8

Konfigurasikan sub-interface ge0/0/1.30 untuk mengelola req ip jaringan 30.30.30.0 /24. Karena tidak ada ketentuan tambahan, cukup ketik perintah dhcp select interface saja agar interface ini menggunakan dynamic allocation.



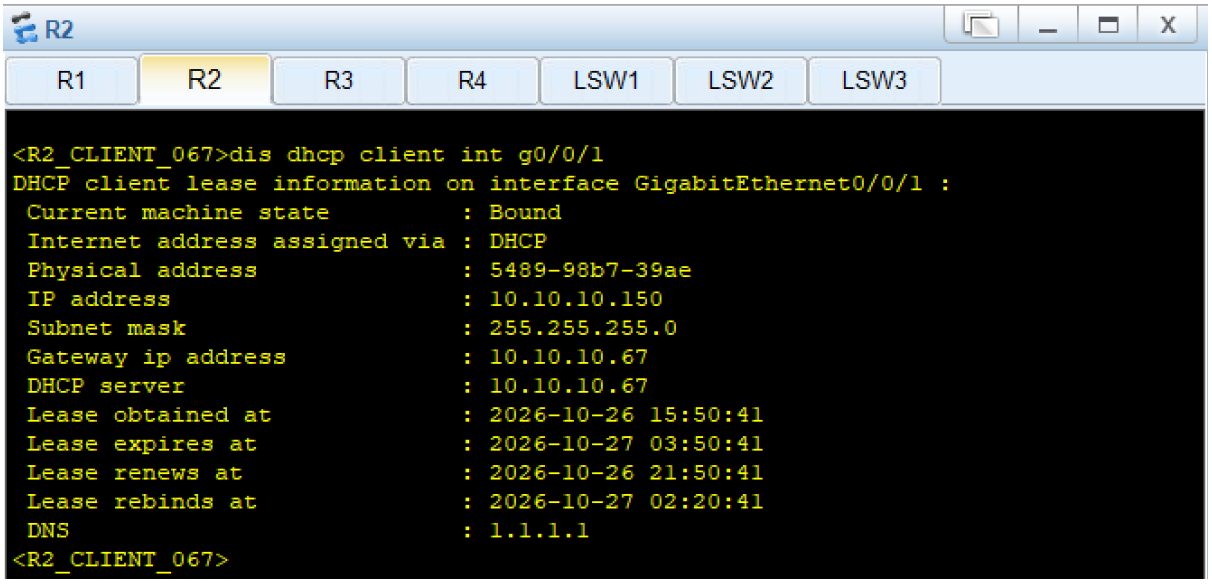
9. Soal 9

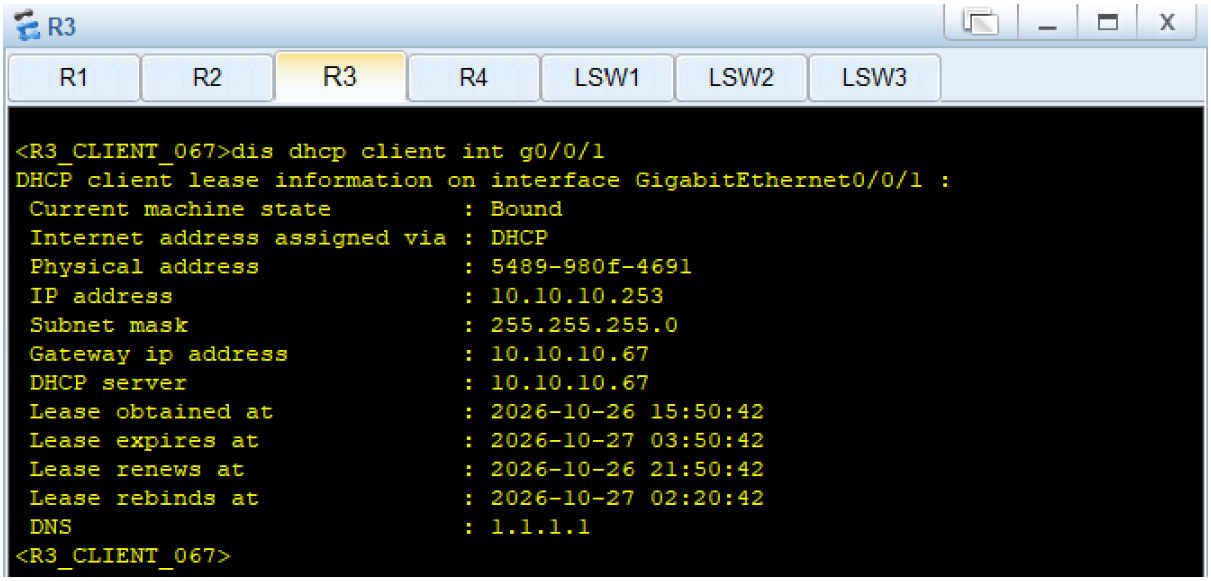
Pemberian label IP dan label VLAN pada setiap perangkat. Untuk mendapatkan IP router client yang sudah diberikan dari server, ketik perintah display ip interface brief, dan cari IP address pada display. Untuk mendapatkan IP PC yang sudah diberikan dari server, ketik perintah ipconfig /renew, dan cari ipv4 address pada display.

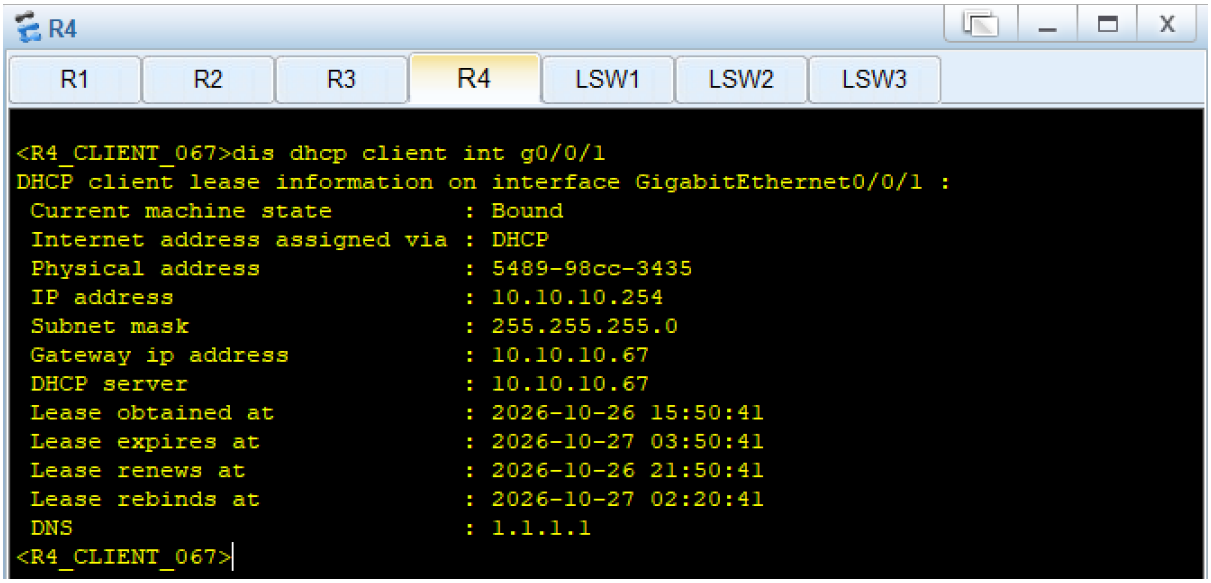


10. Soal 10

Memverifikasi status pelekatan DHCP pada sisi router client menggunakan display dhcp client interface g0/0/1. Hasil eksekusi ini membuktikan router mendapatkan IP Address, subnet, gateway, dan masa sewa secara akurat dari pool R1.



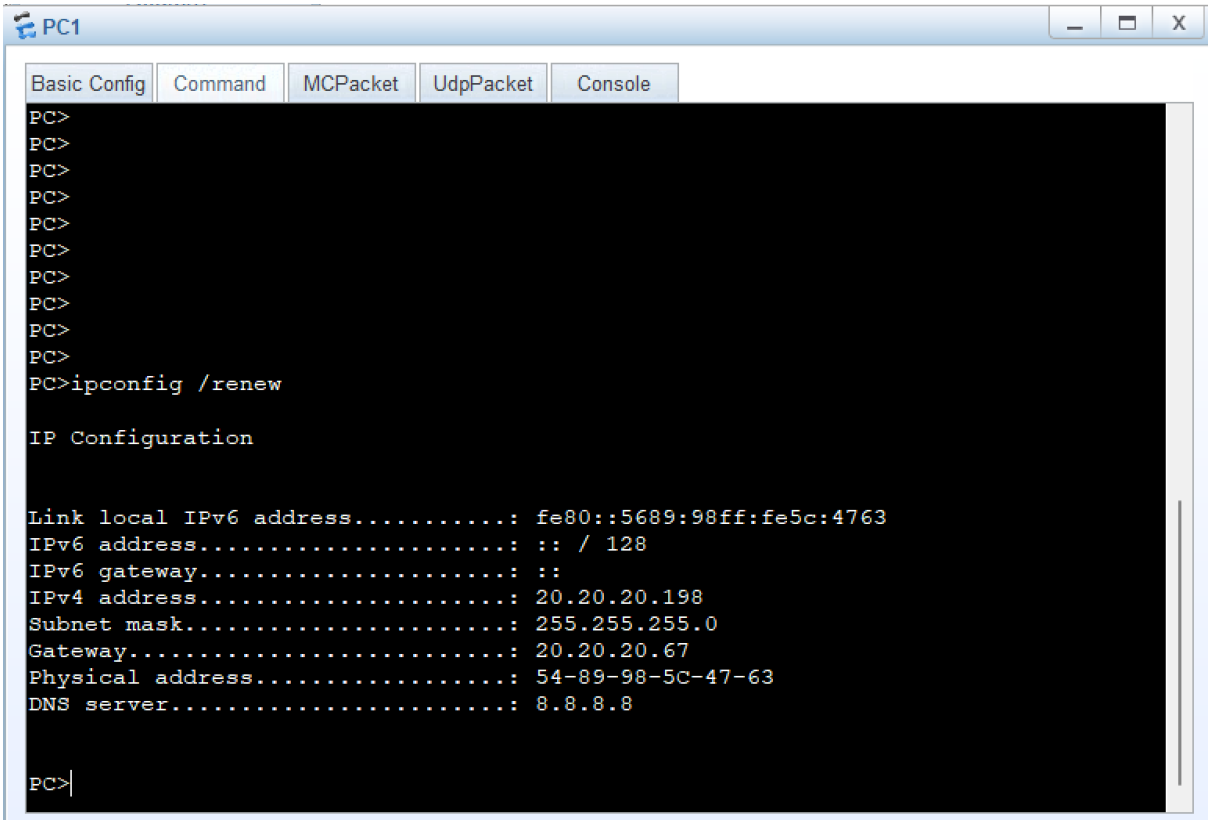


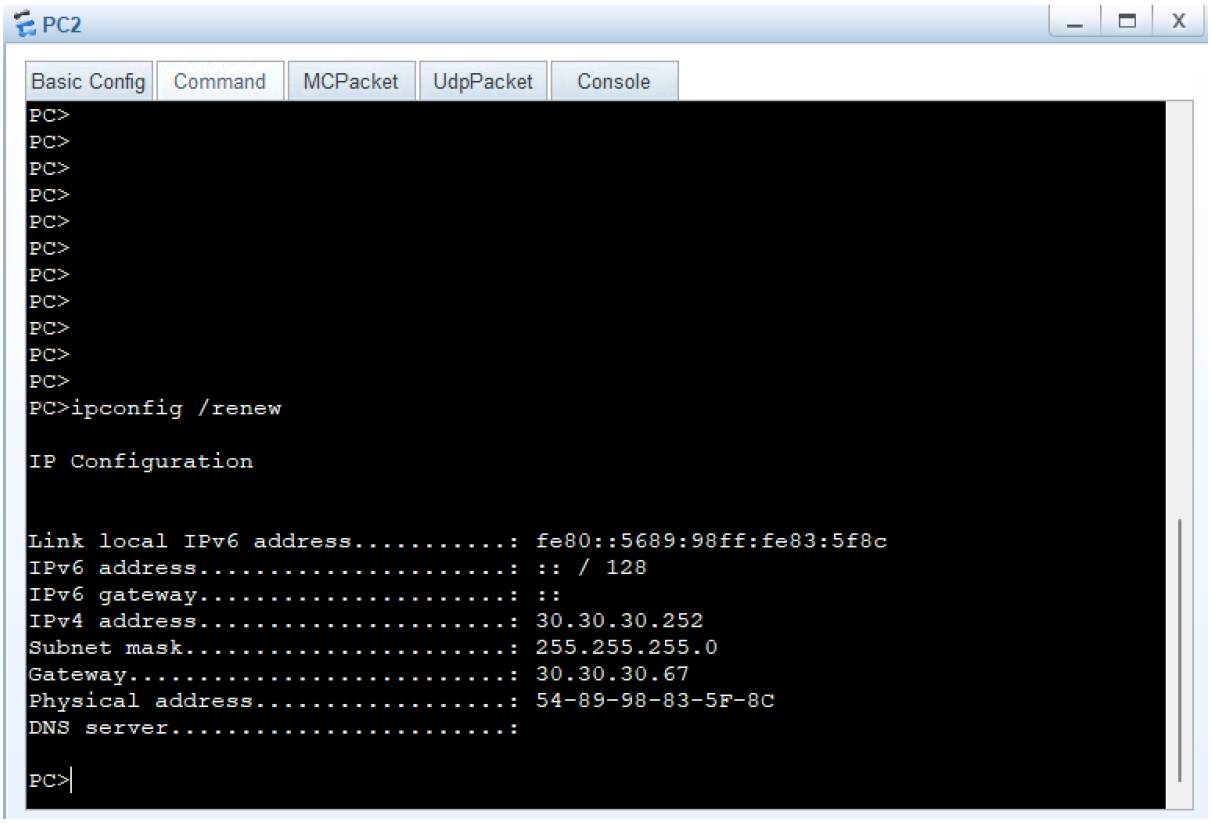


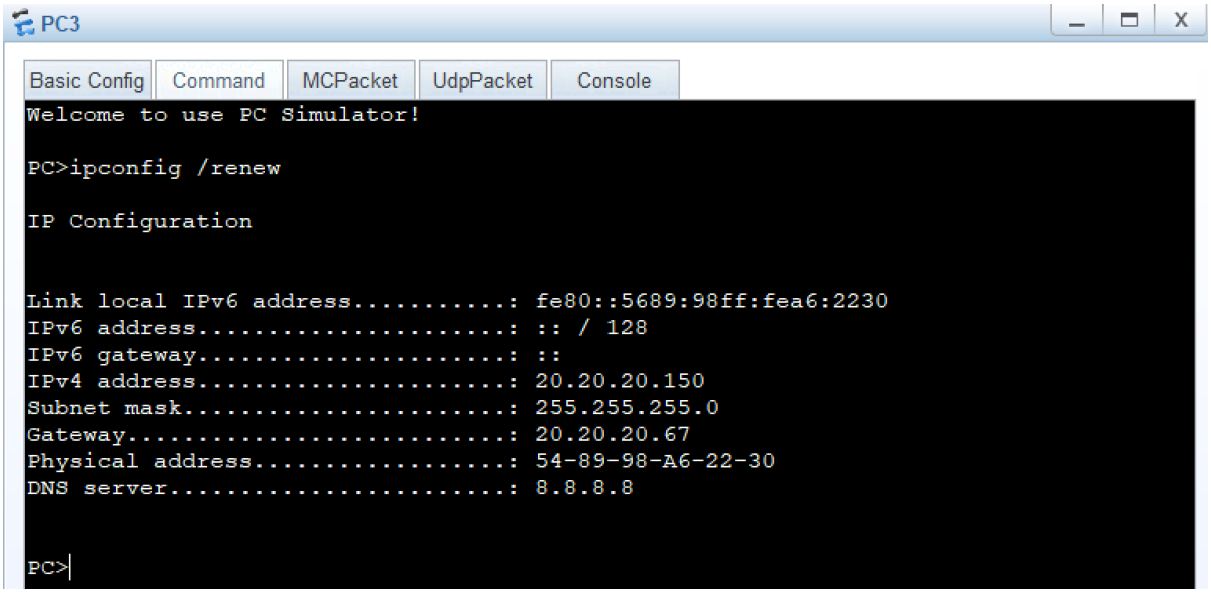
11. Soal 11

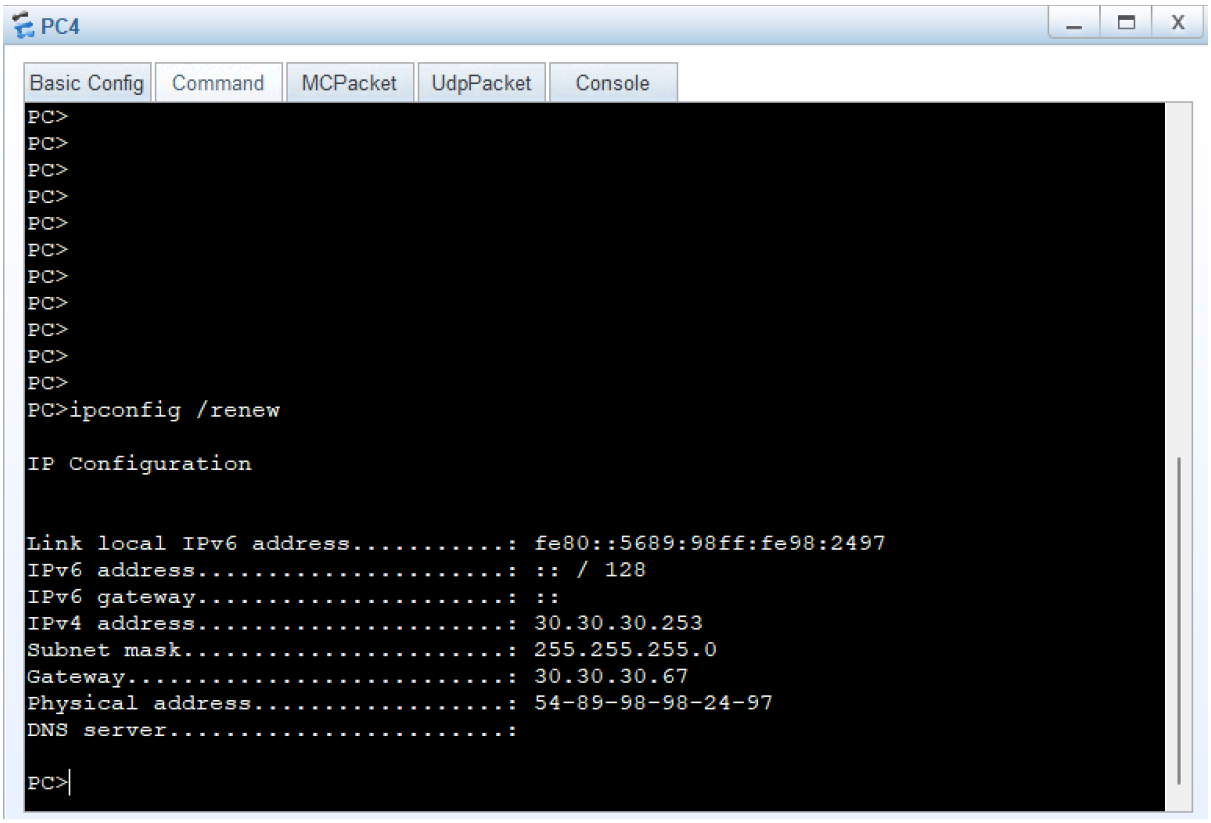
Memaksa pembaruan sewa alamat jaringan pada end-device PC dengan menjalankan perintah ipconfig /renew di jendela command, membuktikan konektivitas layer 2 dari PC, melewati

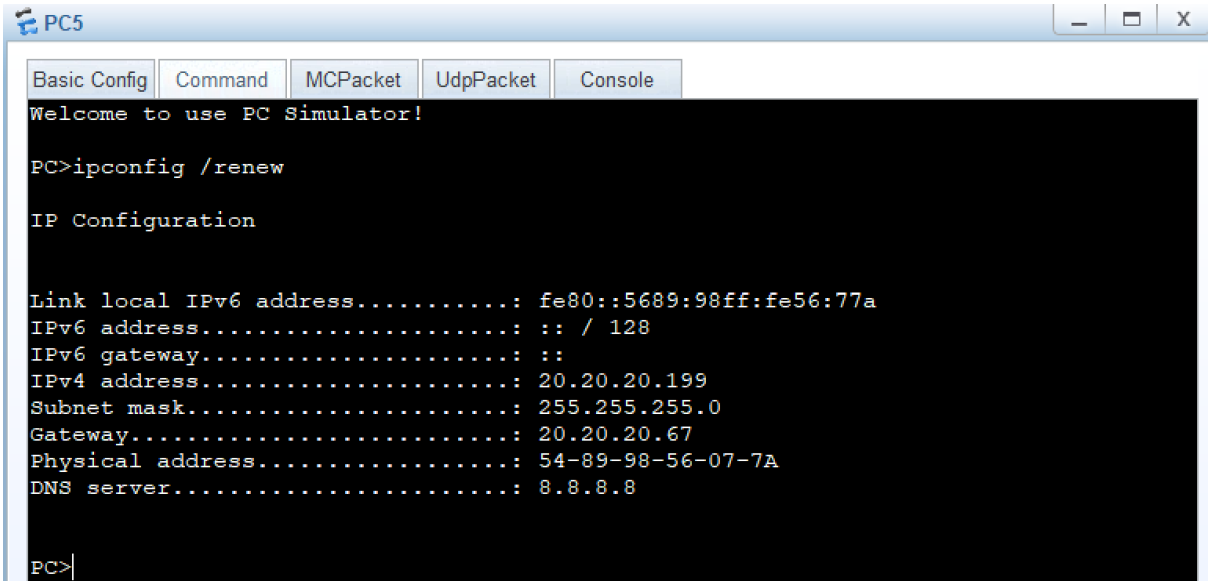
switch, hingga ke DHCP server R1 telah terbentuk.

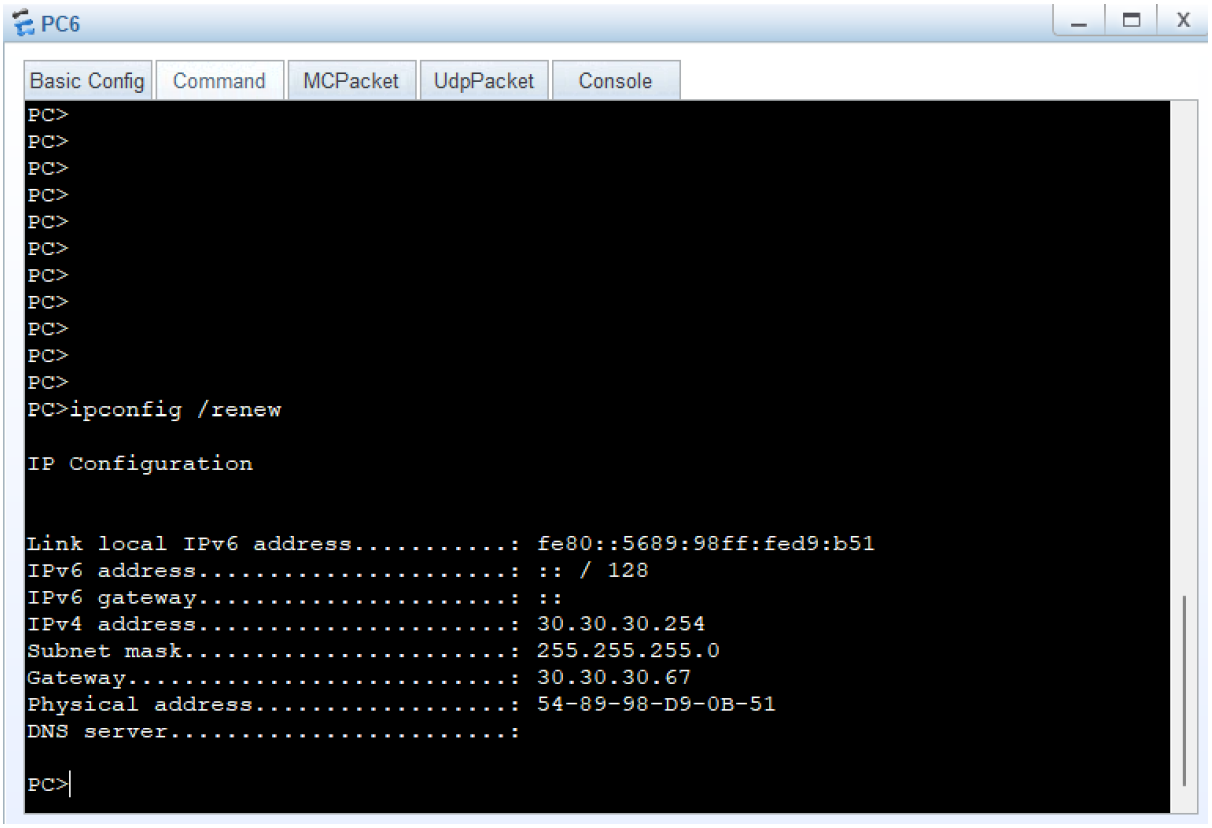





PC3
Basic Config
Command
MCPacket
UdpPacket
Console
Welcome to use PC Simulator!
PC>ipconfig /renew
IP Configuration
Link local IPv6 address.....: fe80::5689:98ff:fea6:2230
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 20.20.20.150
Subnet mask.....: 255.255.255.0
Gateway.....: 20.20.20.67
Physical address.....: 54-89-98-A6-22-30
DNS server.....: 8.8.8.8
PC>

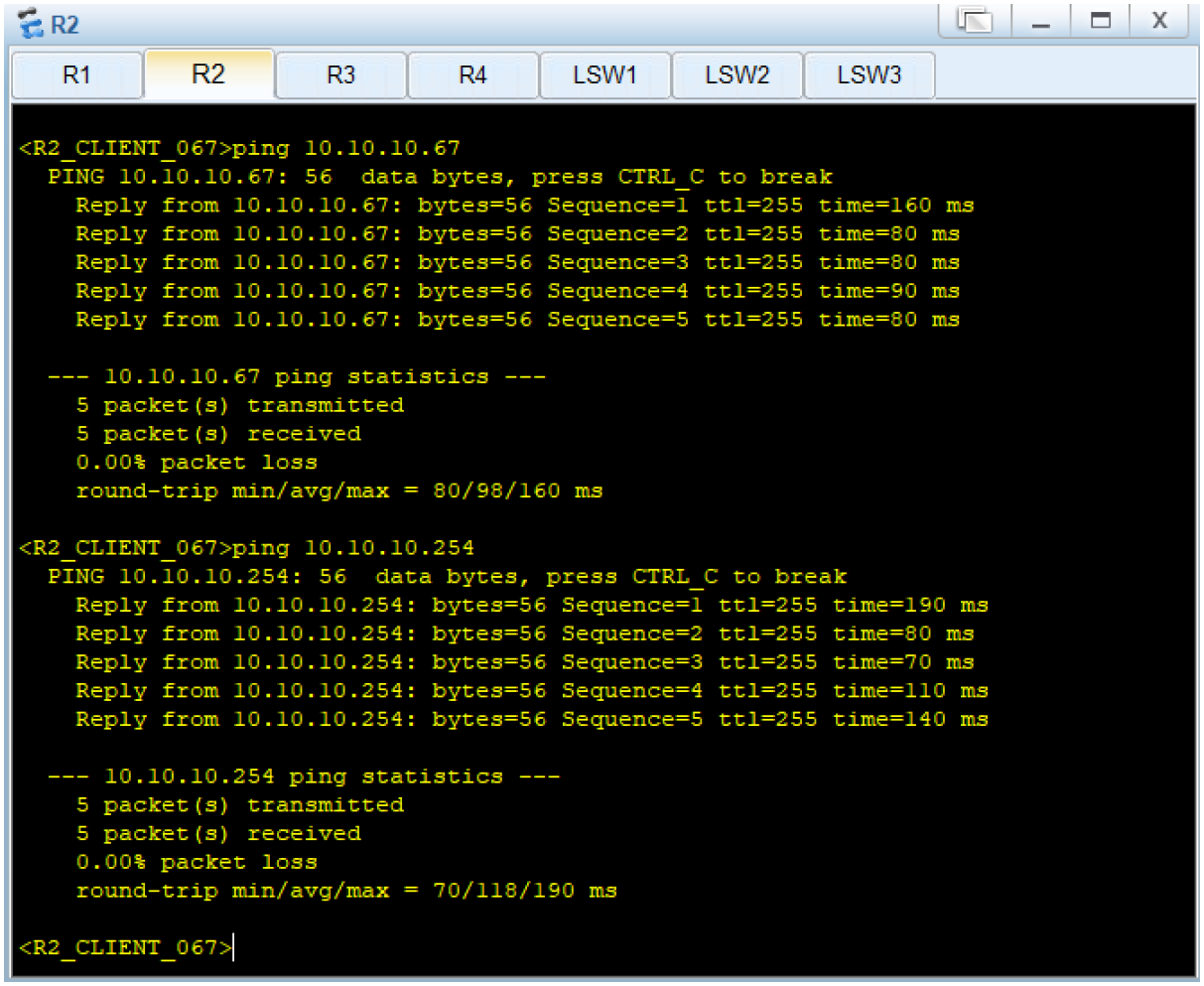

PC4
Basic Config
Command
MCPacket
UdpPacket
Console
PC>
PC>
PC>
PC>
PC>
PC>
PC>
PC>
PC>
PC>
PC>
PC>ipconfig /renew
IP Configuration
Link local IPv6 address.....: fe80::5689:98ff:fe98:2497
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 30.30.30.253
Subnet mask.....: 255.255.255.0
Gateway.....: 30.30.30.67
Physical address.....: 54-89-98-98-24-97
DNS server.....:
PC>

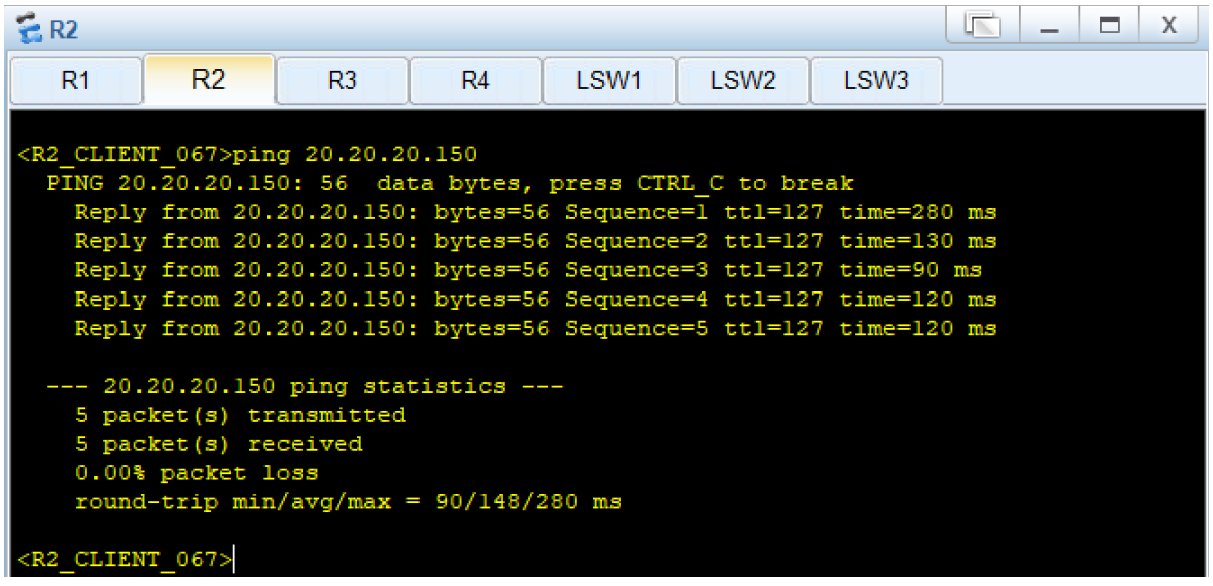




12. Soal 12

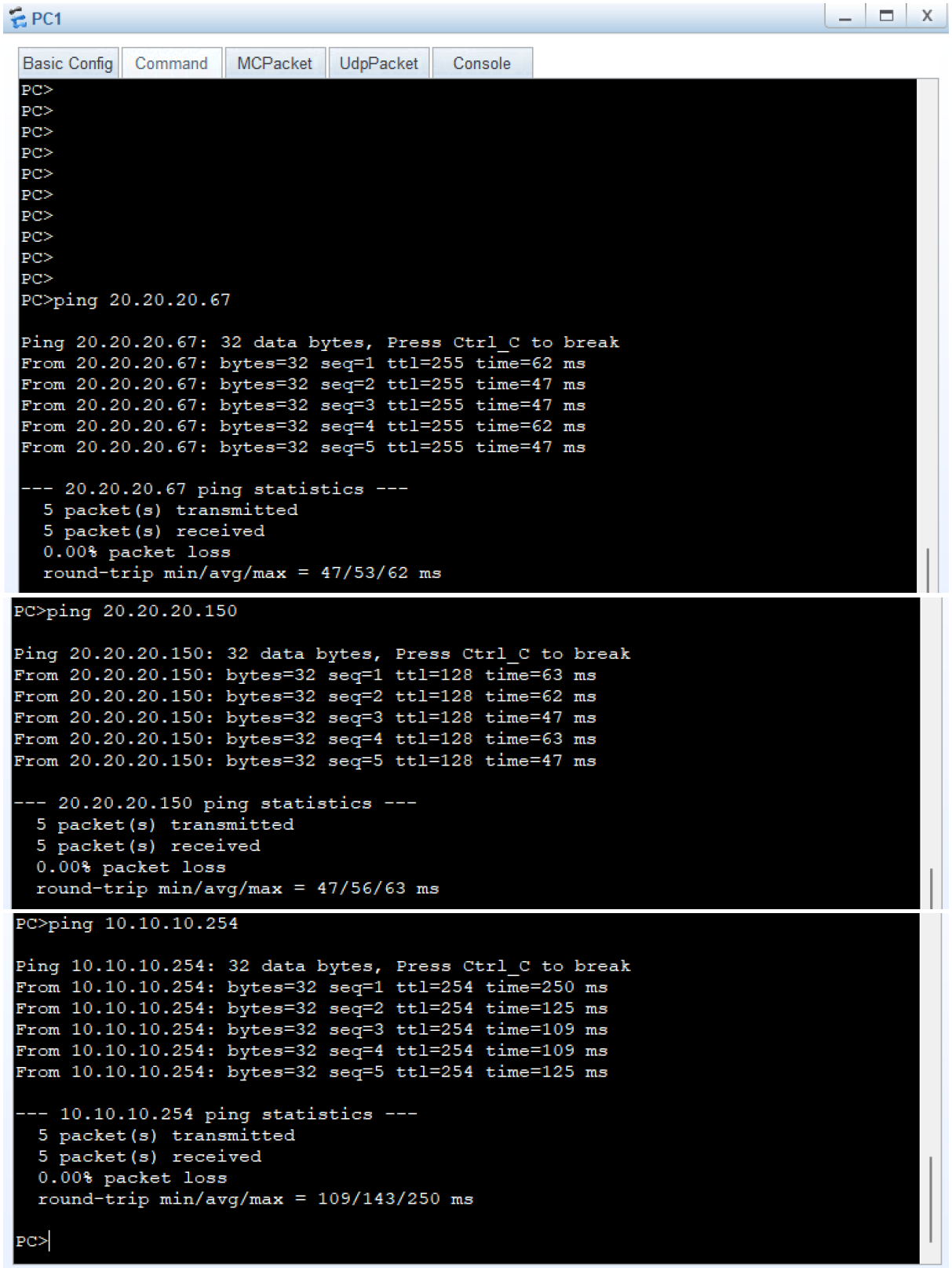
Melakukan uji routing inter-VLAN dari R2 (VLAN 10) dengan melontarkan paket (ping) ke titik gerbang utama R1, lalu ke host internal di jaringan yang sama (R4), dan menyeberang segmen ke host VLAN lain (PC3).





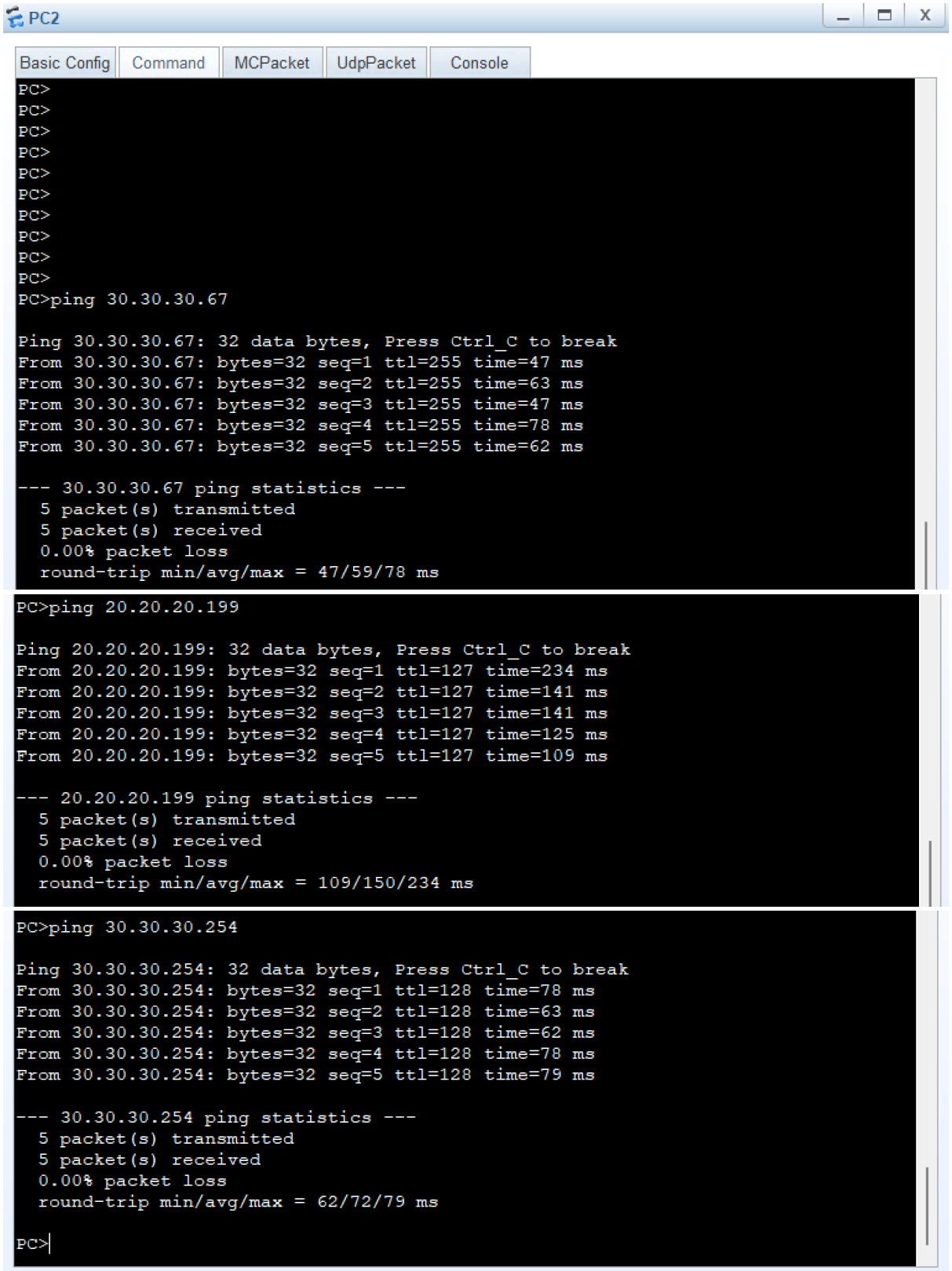
13. Soal 13

Memvalidasi lalu lintas bolak-balik dari PC1 (VLAN 20) menuju IP gateway R1, host di subnet lokal (PC3), dan perangkat beda kelas (R4).



14. Soal 14

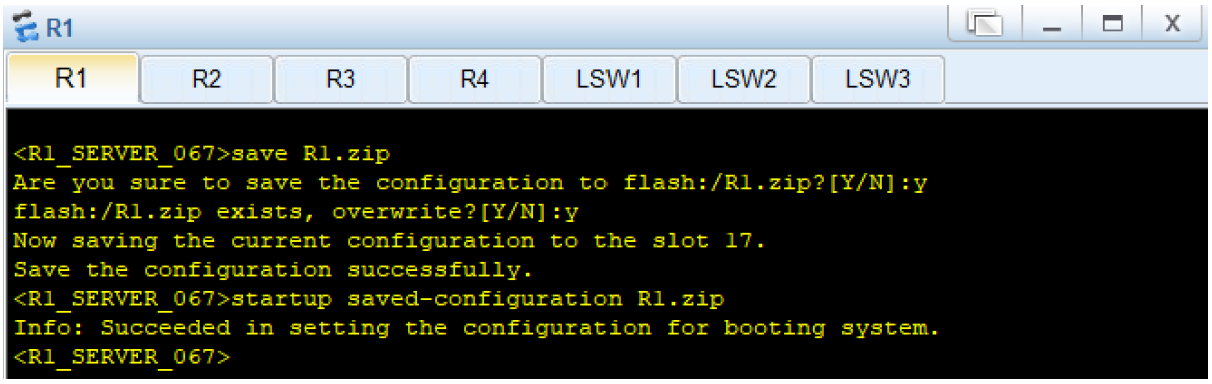
Menguji jangkauan PC2 (VLAN 30) untuk mencapai interface R1, ping anggota sesama VLAN 30 (PC6), serta memastikan tabel routing router mampu meneruskan ping dari PC2 ke PC5 (VLAN 20).

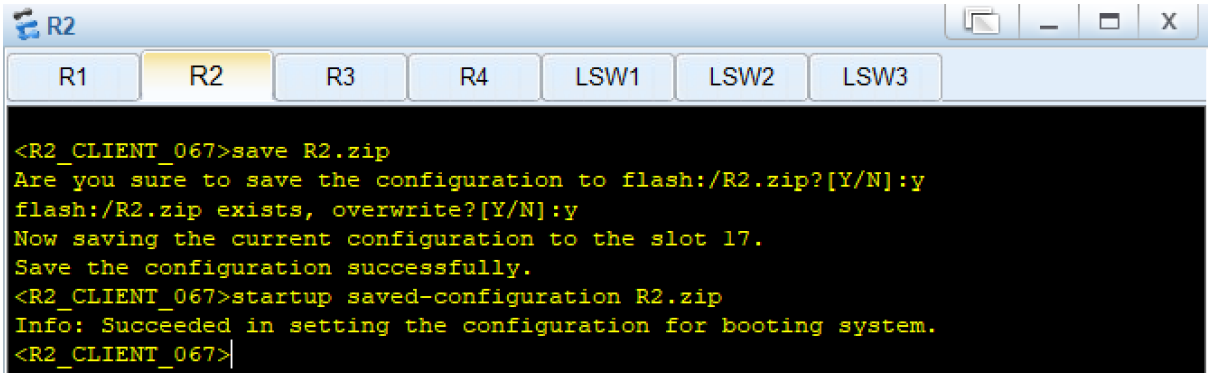


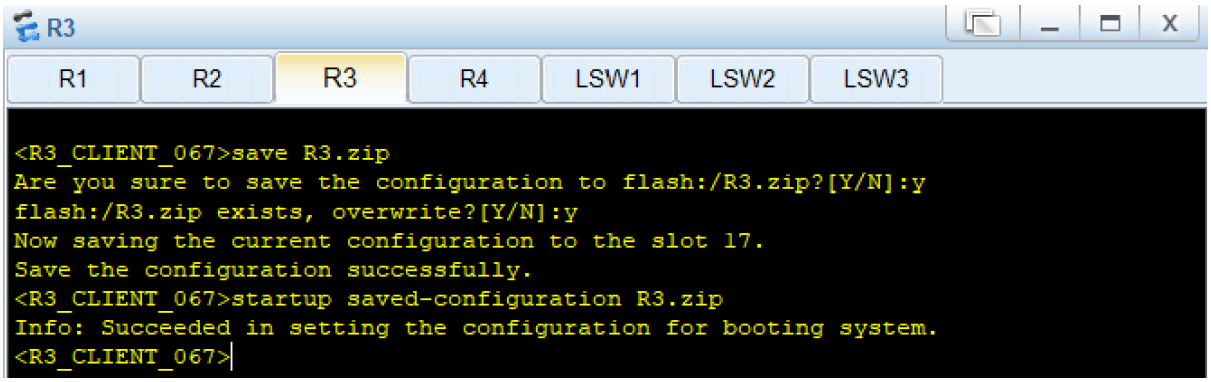
15. Soal 15

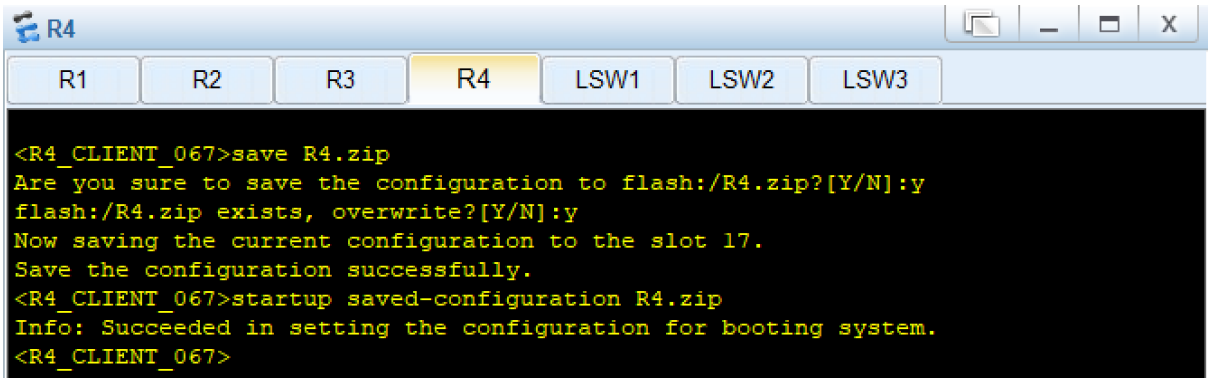
Simpan konfigurasi perangkat dengan perintah save <nama_file>.zip. Setelah itu, startup saved-configuration <nama_file>.zip untuk menjadikan file zip tersebut menjadi konfigurasi

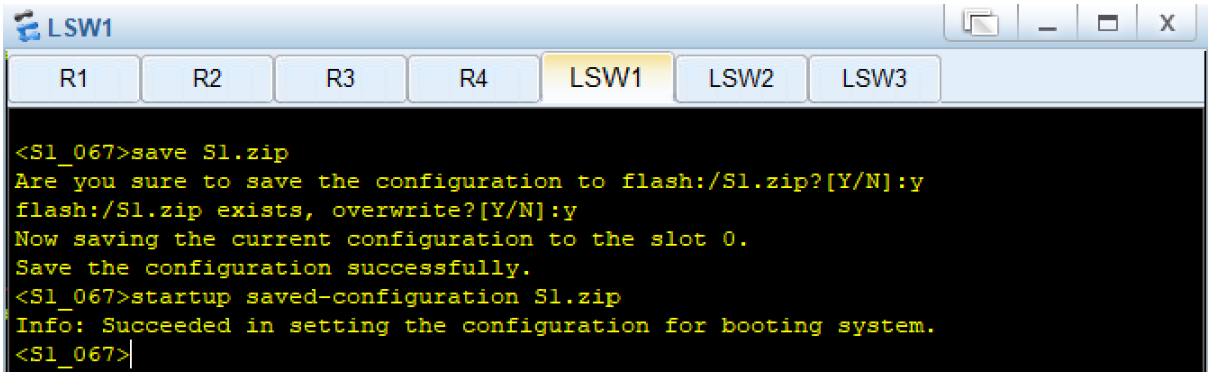
awal saat kembali menyalakan perangkat atau keseluruhan topologi di sesi berikutnya. Lakukan ini kepada setiap perangkat (R1, R2, R3, R4, LSW1, LSW2, dan LSW3).

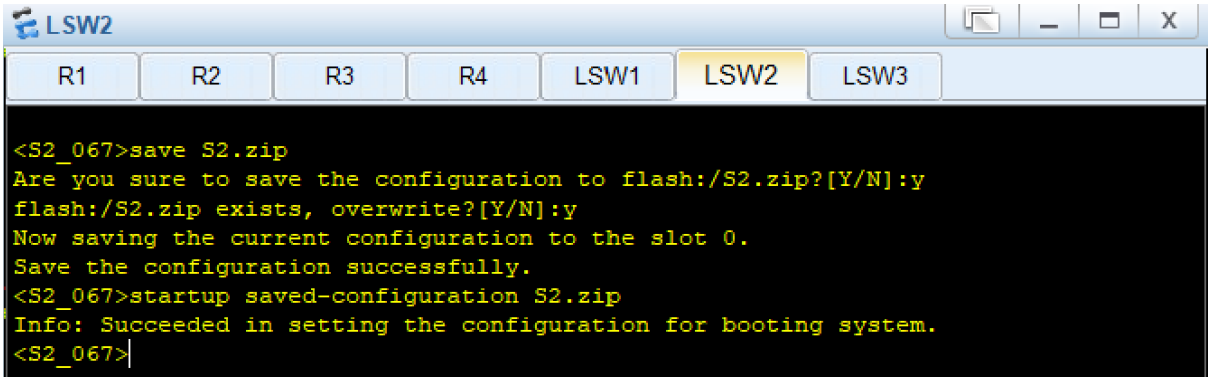


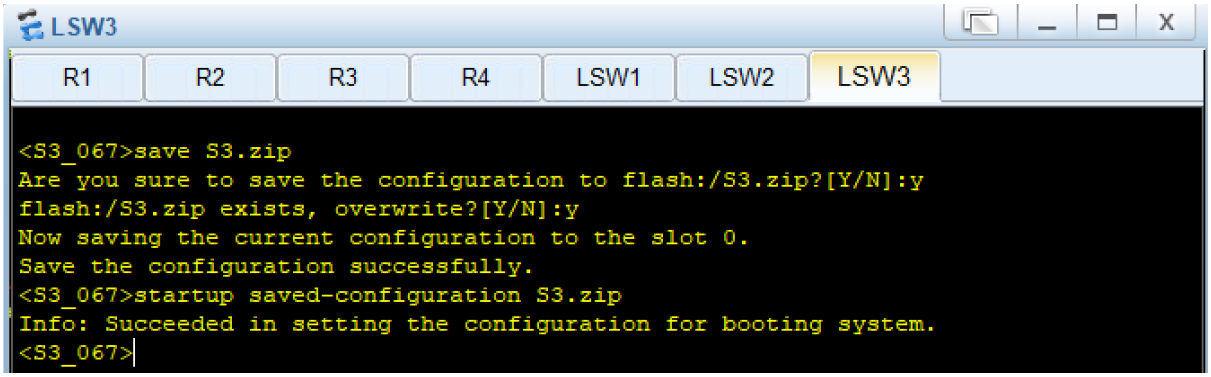







LSW1
R1
R2
R3
R4
LSW1
LSW2
LSW3
<S1_067>save S1.zip
Are you sure to save the configuration to flash:/S1.zip?[Y/N]:y
flash:/S1.zip exists, overwrite?[Y/N]:y
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<S1_067>startup saved-configuration S1.zip
Info: Succeeded in setting the configuration for booting system.
<S1_067>


LSW2
R1
R2
R3
R4
LSW1
LSW2
LSW3
<S2_067>save S2.zip
Are you sure to save the configuration to flash:/S2.zip?[Y/N]:y
flash:/S2.zip exists, overwrite?[Y/N]:y
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<S2_067>startup saved-configuration S2.zip
Info: Succeeded in setting the configuration for booting system.
<S2_067>


LSW3
R1
R2
R3
R4
LSW1
LSW2
LSW3
<S3_067>save S3.zip
Are you sure to save the configuration to flash:/S3.zip?[Y/N]:y
flash:/S3.zip exists, overwrite?[Y/N]:y
Now saving the current configuration to the slot 0.
Save the configuration successfully.
<S3_067>startup saved-configuration S3.zip
Info: Succeeded in setting the configuration for booting system.
<S3_067>